Mickey ch

PHY101-(An Introduction to Physics)

Updated Mid Term Important Solved Mcq's

By Mickey Ch

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1. The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately___

a) 7Hz
b) 14 Hz
c) 28 Hz
d) 57 Hz

2. The SI standard of time is based on is.

a) the daily rotation of the earth
b) the frequency of light emitted by Kr86
c) the yearly revolution of the earth about the sun
d) none of these

3. A net torque applied to a rigid object always tends to produce:

a) rotational equilibrium
b) linear acceleration
c) angular acceleration
d) rotational inertia

4. A nanosecond is_____

a) 109 s
b) 10−9 s
c) 10−10 s
d) 10−10 s

5. To raise the pitch of a certain piano string, the piano tuner:

a) loosens the string
b) tightens the string
c) shortens the string
d) lengthens the string

6. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

a) 12.6 rad/s
b) 1.57 rad/s
c) 2.0 rad/s
d) 6.3 rad/s

7. A force of 5000N is applied outwardly to each end of a 5.0-m long rod with a radius of 34.0 cm and a Young's modulus of 125 × 108 N/m2. The elongation of the rod is:

a) 0.0020mm

b) 0.0040mm
c) 0.14mm
d) 0.55mm

8. In 1866, the U. S. Congress defined the U. S. yard as exactly 3600/3937 international meter. This was done primarily because:

a) length can be measured more accurately in meters than in yards
b) the meter is more stable than the yard
c) this definition relates the common U. S. length units to a more widely used system
d) the members of this Congress were exceptionally intelligent

9. There is no SI base unit for area because:

a) an area has no thickness; hence no physical standard can be built
b) we live in a three (not a two) dimensional world
c) it is impossible to express square feet in terms of meters
d) area can be expressed in terms of square meters

10. A particle oscillating in simple harmonic motion is:

a) never in equilibrium because it is in motion
b) never in equilibrium because there is always a force
c) in equilibrium at the ends of its path because its velocity is zero there
d) in equilibrium at the center of its path because the acceleration is zero there

11. In simple harmonic motion, the restoring force must be proportional to the:

a) amplitude
b) frequency
c) velocity
d) displacement

12. Which of the following weighs about a pound?

a) 0.05 kg
b) 0.5kg
c) 5 kg
d) 50 kg

13. The number of significant figures in 0.00150 is:

a) 2
b) 3
c) 4
d) 5

14. A 160-N child sits on a light swing and is pulled back and held with a horizontal force of 100 N. The magnitude of the tension force of each of the two supporting ropes is:

a) 60N
b) 94N
c) 120N
d) 190N

15. Mercury is a convenient liquid to use in a barometer because:

a) it is a metal
b) it has a high boiling point
c) it expands little with temperature
d) it has a high density

16. 1 mi is equivalent to 1609 m so 55 mph is:

a) 15 m/s
b) 25 m/s
c) 66 m/s
d) 88 m/s

17. A sphere with a radius of 1.7 cm has a surface area of:

a) 2.1 × 10−5 m2
b) 9.1 × 10−4 m2
c) 3.6 × 10−3 m2
d) 0.11 m2

18. A right circular cylinder with a radius of 2.3 cm and a height of 1.4 cm has a total surface area of:

a) 1.7 × 10−3 m2
b) 3.2 × 10−3 m2
c) 2.0 × 10−3 m3
d) 5.3 × 10−3 m2

19. 1 m is equivalent to 3.281 ft. A cube with an edge of 1.5 ft has a volume of:

a) 1.2 × 102 m3
b) 9.6 × 10−2 m3
c) 9.6 × 10−2 m3 C. 10.5m3
d) 9.5 × 10−2 m3

20. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

a) the center of mass
b) the center of gravity

c) the geometrical center
d) the same point

21. The units of the electric field are:

a) J/m
b) J/(C·m)
c) J/C
d) J·C

22. Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

a) 3.14 rad/s2
b) 30 rad/s2
c) 30 rev/s2
d) 50 rev/min2

23. Suppose A = BC, where A has the dimension L/M and C has the dimension L/T. Then B has the dimension:

a) T/M
b) L2/TM
c) TM/L2
d) L2T/M

24. Suppose A = BnCm, where A has dimensions LT, B has dimensions L2T−1, and C has dimensions LT2. Then the exponents n and m have the values:

a) 2/3; 1/3
b) 2; 3
c) 4/5; −1/5
d) 1/5; 3/5

25. A particle moves along the x axis from xi to xf. Of the following values of the initial and final coordinates, which results in a negative displacement?

a) xi =4m, xf =6m
b) xi = −4m, xf = −8m
c) xi = −4m, xf =2m
d) xi = −4m, xf = −2m

26. A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

a) 1.0m/s
b) 1.5m/s

c) 2.0m/s
d) 2.3m/s

27. We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

a) 1 F
b) 1mF
c) 1 μF
d) 100 μF

28. The average speed of a moving object during a given interval of time is always:

a) the magnitude of its average velocity over the interval
b) the distance covered during the time interval divided by the time interval
c) one-half its speed at the end of the interval
d) its acceleration multiplied by the time interval

29. Two automobiles are 150 kilometers apart and traveling toward each other. One automobile is moving at 60km/h and the other is moving at 40km/h mph. In how many hours will they meet?

a) 2.5
b) 2.0
c) 1.75
d) 1.5

30. A car starts from Hither, goes 50 km in a straight line to Yon, immediately turns around, and returns to Hither. The time for this round trip is 2 hours. The magnitude of the average velocity of the car for this round trip is:

a) 0
b) 50 km/hr
c) 100 km/hr
d) 200 km/hr

31. An object moving in a circle at constant speed

a) must have only one force acting on it
b) is not accelerating
c) is held to its path by centrifugal force
d) has an acceleration of constant magnitude

32. A plane traveling north at 200m/s turns and then travels south at 200m/s. The change in its velocity is:

a) 400m/s north

b) **400m/s south**
c) zero
d) 200m/s south

33. The wavelength of red light is 700 nm. Its frequency is __________________.

a) 4.30 * 104 Hertz
b) 4.30 * 103 Hertz
c) 4.30 * 105 Hertz
d) 4.30 * 102 Hertz

34. Which of the following statements is NOT TRUE about electromagnetic waves?

a) Electromagnetic waves satisfy the Maswell's Equation.
b) Electromagnetic waves can not travel through space.
c) The receptions of electromagnetic waves require an antenna.
d) The electromagnetic radiation from a burning candle is unpolarized

35. At time t = 0 a car has a velocity of 16 m/s. It slows down with an acceleration given by −0.50t, in m/s2 for t in seconds. It stops at t =

a) 64 s
b) 32 s
c) 16 s
d) 8.0 s

36. The coordinate of a particle in meters is given by x(t)= 16t − 3.0t3, where the time t is in

The coordinate of a particle in meters is given by x(t)= 16t − 3.0t3, where the time t is in
seconds. The particle is momentarily at rest at t =

a) 0.75 s
b) 1.3s
c) 5.3s
d) 7.3s

37. A drag racing car starts from rest at t = 0 and moves along a straight line with velocity given by v = bt2, where b is a constant. The expression for the distance traveled by this car from its position at t = 0 is:

a) bt3
b) bt3/3
c) 4bt2
d) 3bt2

38. A ball rolls up a slope. At the end of three seconds its velocity is 20 cm/s; at the end of eight seconds its velocity is 0. What is the average acceleration from the third to the eighth second?

a) 2.5cm/s2

b) 4.0cm/s2
c) 5.0cm/s2
d) 6.0cm/s2

39. The velocity of an object is given as a function of time by v =4t − 3t2, where v is in m/s and The velocity of an object is given as a function of time by v =4t − 3t2, where v is in m/s and t is in seconds. Its average velocity over the interval from t =0 to t =2s:

a) is 0
b) is −2m/s
c) is 2m/s
d) is −4m/s

40. The coordinate of an object is given as a function of time by x =4t2−3t3, where x is in meters The coordinate of an object is given as a function of time by x =4t2−3t3, where x is in meters and t is in seconds. Its average acceleration over the interval from t =0 to t = 2 s is:

a) −4m/s2
b) 4m/s2
c) −10m/s2
d) 10m/s241

42. Each of four particles move along an x axis. Their coordinates (in meters) as functions of time (in seconds) are given by

particle 1: x(t)=3.5 − 2.7t3

particle 2: x(t)=3.5+ 2.7t3

particle 3: x(t)=3.5+ 2.7t2

particle 4: x(t)=3.5 − 3.4t − 2.7t2

Which of these particles is speeding up for t> 0?

a) All four
b) Only 1
c) Only 2 and 3
d) Only 2, 3, and 4

43. 1 mi is equivalent to 1609 m so 55 mph is:

a) 15 m/s
b) 25 m/s
c) 66 m/s
d) 88 m/s

44. Radio waves and light waves are ____________________________.

a) Longitudinal waves
b) Transverse waves

c) **Electromagnetic and transverse both**
d) Electromagnetic and longitudinal both

45. Wien's Law states that, lmax = ________ K.

a) 2.90 * 10-3 Hertz
b) 2.90 * 10-3 s
c) 2.90 * 10-3 kg
d) **2.90 * 10-3 m**

46. For a body to be in equilibrium under the combined action of several forces:

a) all the forces must be applied at the same point
b) all of the forces form pairs of equal and opposite forces
c) **any two of these forces must be balanced by a third force**
d) the sum of the torques about any point must equal zero

47. As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the

angular momentum around the origin is:

a) **6.0kg · m2/s**
b) 9.0kg · m2/s
c) 11 kg · m2/s
d) 14 kg · m2/s

48. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

a) **2.0 rad/s**
b) 12.6 rad/s
c) 1.57 rad/s
d) 6.3 rad/s

49. Interference of light is evidence that:

a) the speed of light is very large
b) light is a transverse wave
c) **light is a wave phenomenon**
d) light is electromagnetic in character

50. Fahrenheit and Kelvin scales agree numerically at a reading of:

a) **-40**
b) 0
c) 273
d) 574

51. According to the theory of relativity:

a) moving clocks run fast
b) energy is not conserved in high speed collisions
c) the speed of light must be measured relative to the ether
d) none of the above are true

52. Of the following situations, which one is impossible?

a) A body having velocity east and acceleration east
b) A body having velocity east and acceleration west
c) A body having zero velocity and non-zero acceleration
d) A body having constant velocity and variable acceleration

53. Throughout a time interval, while the speed of a particle increases as it moves along the x axis, its velocity and acceleration might be:

a) positive and negative, respectively
b) negative and positive, respectively
c) negative and negative, respectively
d) negative and zero, respectively

54. A particle moves on the x axis. When its acceleration is positive and increasing:

a) its velocity must be positive
b) its velocity must be negative
c) it must be slowing down
d) none of the above must be true

55. The position y of a particle moving along the y axis depends on the time t according to the equation y = at − bt2. The dimensions of the quantities a and b are respectively:

a) L2/T, L3/T2
b) L/T2,L2/T
c) L/T, L/T2
d) L3/T, T2/L

56. Over a short interval near time t = 0 the coordinate of an automobile in meters is given by x(t)= 27t − 4.0t3, where t is in seconds. At the end of 1.0 s the acceleration of the auto is:

a) 27 m/s2
b) 4.0 m/s2
c) −4.0 m/s2
d) −24 m/s2

57. Over a short interval, starting at time t = 0, the coordinate of an automobile in meters is given by x(t)= 27t − 4.0t3, where t is in seconds. The magnitudes of the initial (at t = 0) velocity and acceleration of the auto respectively are:

a) 0; 12 m/s2
b) 0; 24 m/s2
c) 27 m/s; 0
d) 27 m/s; 12 m/s2

58. At time t = 0 a car has a velocity of 16 m/s. It slows down with an acceleration given by −0.50t, in m/s2 for t in seconds. It stops at t =

a) 64 s
b) 32 s
c) 16 s
d) 8.0s

59. Starting at time t = 0, an object moves along a straight line. Its coordinate in meters is given by x(t)= 75t − 1.0t3, where t is in seconds. When it momentarily stops its acceleration is:

a) 0
b) −73 m/s2
c) −30 m/s2
d) −9.8 m/s2

60. A car, initially at rest, travels 20 m in 4 s along a straight line with constant acceleration. The acceleration of the car is:

a) 0.4m/s2
b) 1.3m/s2
c) 2.5m/s2
d) 4.9m/s2

61. A car starts from rest and goes down a slope with a constant acceleration of 5 m/s2. After 5 s the car reaches the bottom of the hill. Its speed at the bottom of the hill, in meters per second, is:

a) 1
b) 12.5
c) 25
d) 50

62. A car moving with an initial velocity of 25 m/s north has a constant acceleration of 3 m/s2 south. After 6 seconds its velocity will be:

a) 7 m/s north
b) 7 m/s south
c) 43 m/s north
d) 20 m/s north

63. An object with an initial velocity of 12 m/s west experiences a constant acceleration of 4 m/s2 west for 3 seconds. During this time the object travels a distance of:

a) 12 m
b) 24 m
c) 36 m
d) 54 m

64. At a stop light, a truck traveling at 15 m/s passes a car as it starts from rest. The truck travels at constant velocity and the car accelerates at 3 m/s2. How much time does the car take to catch up to the truck?

a) 5 s
b) 10 s
c) 15 s
d) 20 s

65. A ball is in free fall. Its acceleration is:

a) downward during both ascent and descent
b) downward during ascent and upward during descent
c) upward during ascent and downward during descent
d) upward during both ascent and descent

66. Which of the following statements is NOT TRUE about electromagnetic waves?

a) Electromagnetic waves satisfy the Maswell's Equation.
b) Electromagnetic waves can not travel through space.
c) The receptions of electromagnetic waves require an antenna.
d) The electromagnetic radiation from a burning candle is unpolarized

67. How fast should you move away from a 6.0 × 1014 Hz light source to observe waves with a frequency of 4.0 × 1014 Hz?

a) 20c
b) 38c
c) 45c
d) 51c

68. The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

a) Magnetic momentum
b) Energy
c) Orbital angular momentum
d) Spin angular momentum

69. As the wavelength of a wave in a uniform medium increases, its speed will _____.

a) Decrease
b) Increase
c) Remain the same
d) None of these

70. Velocity is defined as:

a) rate of change of position with time
b) position divided by time
c) rate of change of acceleration with time
d) a speeding up or slowing down

71. Acceleration is defined as:

a) rate of change of position with time
b) speed divided by time
c) rate of change of velocity with time
d) a speeding up or slowing down

72. Which of the following is a scalar quantity?

a) Speed
b) Velocity
c) Displacement
d) Acceleration

73. Which of the following is NOT an example of accelerated motion?

a) Vertical component of projectile motion
b) Circular motion at constant speed
c) A swinging pendulum
d) Horizontal component of projectile motion

74. A particle goes from $x = -2m$, $y = 3m$, $z = 1m$ to $x = 3m$, $y = -1m$, $z = 4m$. Its displacement is:

a) $(1m)\hat{i} + (2m)\hat{j} + (5m)\hat{k}$
b) $(5m)\hat{i} - (4m)\hat{j} + (3m)\hat{k}$
c) $-(5m)\hat{i} + (4m)\hat{j} - (3m)\hat{k}$
d) $-(1m)\hat{i} - (2m)\hat{j} - (5m)\hat{k}$

75. A jet plane in straight horizontal flight passes over your head. When it is

directly above you, the sound seems to come from a point behind the plane in a direction 30° from the vertical. The speed of the plane is:

a) the same as the speed of sound
b) half the speed of sound
c) three-fifths the speed of sound

d) 0.866 times the speed of sound

76. Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

a) higher frequency and a longer wavelength than before
b) lower frequency and a shorter wavelength than before
c) higher frequency and a shorter wavelength than before
d) lower frequency and a longer wavelength than before

77. The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

a) 7Hz
b) 14 Hz
c) 28 Hz
d) 57 Hz

78. An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

a) 12.6 rad/s
b) 1.57 rad/s
c) 2.0 rad/s
d) 6.3 rad/s

79. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

a) the center of mass
b) the center of gravity
c) the geometrical center
d) the same point

80. A baseball is thrown vertically into the air. The acceleration of the ball at its highest point is:

a) zero
b) g,down
c) g,up
d) 2g,down

81. A freely falling body has a constant acceleration of 9.8 m/s2. This means that:

a) the body falls 9.8 m during each second
b) the body falls 9.8 m during the first second only
c) the speed of the body increases by 9.8 m/s during each second

d) the acceleration of the body increases by 9.8 m/s2 during each second

82. An object is thrown straight up from ground level with a speed of 50 m/s. If g = 10 m/s2 its distance above ground level 1.0 s later is:

a) 40 m
b) 45 m
c) 50 m
d) 55 m

83. An object is thrown straight up from ground level with a speed of 50 m/s. If g = 10 m/s2 its distance above ground level 6.0 s later is:

a) 0.00 m
b) 270 m
c) 480 m
d) none of these

84. At a location where g =9.80 m/s2, an object is thrown vertically down with an initial speed of 1.00 m/s. After 5.00 s the object will have traveled:

a) 125 m
b) 127.5m
c) 245 m
d) 250 m

85. A feather, initially at rest, is released in a vacuum 12 m above the surface of the earth. Which of the following statements is correct?

a) The maximum velocity of the feather is 9.8 m/s
b) The acceleration of the feather decreases until terminal velocity is reached
c) The acceleration of the feather remains constant during the fall
d) The acceleration of the feather increases during the fall

86. A heavy ball falls freely, starting from rest. Between the third and fourth second of time it travels a distance of:

a) 4.9m
b) 9.8m
c) 29.4m
d) 34.3m

87. A stone is thrown vertically upward with an initial speed of 19.5 m/s. It will rise to a maximum height of:

a) 4.9m
b) 9.8m
c) 19.4m

d) 38.8m

88. A plane traveling north at 200m/s turns and then travels south at 200m/s.

The change in its velocity is:

a) zero
b) 200m/s north
c) 300m/s north
d) 400m/s south

89. The velocity of a projectile equals its initial velocity added to:

a) a constant horizontal velocity
b) a constant vertical velocity
c) a constantly increasing horizontal velocity
d) a constantly increasing downward velocity

90. A stone thrown from the top of a tall building follows a path that is:

a) circular
b) made of two straight line segments
c) hyperbolic
d) parabolic

91. The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

a) 7Hz
b) 14 Hz
c) 28Hz
d) 57Hz

92. Identical guns fire identical bullets horizontally at the same speed from the same height above level planes, one on the Earth and one on the Moon. Which of the following three statements is/are true?

I. The horizontal distance traveled by the bullet is greater for the Moon.

II. The flight time is less for the bullet on the Earth.

III. The velocity of the bullets at impact are the same.

a) III only
b) I and II only
c) I and III only
d) II and III only

93. An object has a constant acceleration of 3 m/s2. The coordinate versus time graph for this object has a slope:

a) that increases with time
b) that is constant
c) that decreases with time
d) of 3 m/s

94. A vector of magnitude 3 CANNOT be added to a vector of magnitude 4 so that the magnitude of the resultant is:

a) zero
b) 1
c) 3
d) 5

95. A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of this sum might be:

a) zero
b) 3
c) 12
d) 47

96. A certain vector in the xy plane has an x component of 4m and a y component of 10m. It is then rotated in the xy plane so its x component is doubled. Its new y component is about:

a) 20m
b) 7.2m
c) 5.0m
d) 4.5m

97. Two vectors have magnitudes of 10m and 15 m. The angle between them when they are drawn with their tails at the same point is 65◦. The component of the longer vector along the line of the shorter is:

a) 0
b) 4.2m
c) 6.3m
d) 9.1m

98. If the magnitude of the sum of two vectors is less than the magnitude of either vector, then:

a) the scalar product of the vectors must be negative
b) the scalar product of the vectors must be positive
c) the vectors must be parallel and in opposite directions
d) the vectors must be parallel and in the same direction

99. Two vectors have magnitudes of 10m and 15 m. The angle between them when they are drawn with their tails at the same point is 65◦. The component of the longer vector along the line perpendicular to the shorter vector, in the plane of the vectors, is:

a) 0
b) 4.2m
c) 9.1m
d) 14m

100. Which of the following is NOT an example of accelerated motion?

a) Vertical component of projectile motion
b) Circular motion at constant speed
c) Earth's motion about sun
d) Horizontal component of projectile motion

101. The velocity of a projectile equals its initial velocity added to:

a) a constant horizontal velocity
b) a constant vertical velocity
c) a constantly increasing horizontal velocity
d) a constantly increasing downward velocity

102. A bullet shot horizontally from a gun:

a) strikes the ground much later than one dropped vertically from the same point at the same instant
b) never strikes the ground
c) strikes the ground at approximately the same time as one dropped vertically from the same point at the same instant
d) travels in a straight line

103. A bomber flying in level flight with constant velocity releases a bomb before it is over the target. Neglecting air resistance, which one of the following is NOT true?

a) The bomber is over the target when the bomb strikes
b) The acceleration of the bomb is constant
c) The horizontal velocity of the plane equals the vertical velocity of the bomb when it hits the target
d) The bomb travels in a curved path

104. An object is shot from the back of a railroad flatcar moving at 40km/h on a straight horizontal road. The launcher is aimed upward, perpendicular to the bed of the flatcar. The object falls:

a) in front of the flatcar
b) behind the flatcar
c) on the flatcar
d) either behind or in front of the flatcar, depending on the initial speed of the object

105. A stone is thrown outward from the top of a 59.4-m high cliff with an upward velocity compo-nent of 19.5m/s. How long is stone in the air?

a) 4.00 s
b) 5.00 s
c) 6.00 s
d) 7.00 s

106. A projectile is fired from ground level over level ground with an initial velocity that has a vertical component of 20m/s and a horizontal component of 30m/s. Using g =10m/s2, the distance from launching to landing points is:

a) 40m
b) 60m
c) 80m
d) 120m

107. An airplane makes a gradual 90◦ turn while flying at a constant speed of 200m/s. The process takes 20.0 seconds to complete. For this turn the magnitude of the average acceleration of the plane is:

a) zero
b) 40m/s2
c) 20m/s2
d) 14m/s2

108. A stone is tied to a string and whirled at constant speed in a horizontal circle. The speed is then doubled without changing the length of the string. Afterward the magnitude of the acceleration of the stone is:

a) the same
b) twice as great
c) four times as great
d) half as great

109. A stone is tied to a 0.50-m string and whirled at a constant speed of 4.0m/s in a vertical circle. Its acceleration at the bottom of the circle is:

a) 9.8m/s2,up
b) 9.8m/s2,down
c) 8.0m/s2,up
d) 32m/s2,up

110. A girl jogs around a horizontal circle with a constant speed. She travels one fourth of a revolution, a distance of 25m along the circumference of the circle, in 5.0 s. The magnitude of her acceleration is:

a) 0.31m/s2
b) 1.3m/s2
c) 1.6m/s2
d) 3.9m/s2

111. A Ferris wheel with a radius of 8.0m makes 1 revolution every 10 s. When a passenger is at the top, essentially a diameter above the ground, he releases a ball. How far from the point on the ground directly under the release point does the ball land?

a) 0
b) 1.0m
c) 8.0m
d) 9.1m

112. Two projectiles are in flight at the same time. The acceleration of one relative to the other:

a) is always 9.8m/s2
b) can be as large as 19.8m/s2
c) can be horizontal
d) is zero

113. An example of an inertial reference frame is:

a) any reference frame that is not accelerating
b) a frame attached to a particle on which there are no forces
c) any reference frame that is at rest
d) a reference frame attached to the center of the universe

114. In SI units a force is numerically equal to the

a) velocity of the standard kilogram
b) speed of the standard kilogram
c) velocity of any object
d) acceleration of the standard kilogram

115. The standard 1-kg mass is attached to a compressed spring and the spring is released. If the mass initially has an acceleration of 5.6m/s2, the force of the spring has a magnitude of:

a) 2.8N
b) 5.6N
c) 11.2N
d) 0

116. Acceleration is always in the direction:

a) of the displacement
b) of the initial velocity
c) of the final velocity
d) of the net force

117. The term “mass” refers to the same physical concept as:

a) weight

b) inertia
c) force
d) acceleration

118. When a certain force is applied to the standard kilogram its acceleration is 5.0m/s2. When the same force is applied to another object its acceleration is one-fifth as much. The mass of the object is:

a) 0.2kg
b) 0.5kg
c) 1.0kg
d) 5.0kg

119. The mass of a body:

a) is slightly different at different places on Earth
b) is a vector
c) is independent of the free-fall acceleration
d) is the same for all bodies of the same volume

120. A feather and a lead ball are dropped from rest in vacuum on the Moon. The acceleration of the feather is:

a) more than that of the lead ball
b) the same as that of the lead ball
c) less than that of the lead ball
d) 9.8m/s2

121. A car travels east at constant velocity. The net force on the car is:

a) east
b) west
c) up
d) zero

122. A 6-kg object is moving south. A net force of 12N north on it results in the object having an acceleration of:

a) 2m/s2, north
b) 2m/s2, south
c) 6m/s2, north
d) 18m/s2, north

123. A ball with a weight of 1.5N is thrown at an angle of 30◦ above the horizontal with an initial speed of 12m/s. At its highest point, the net force on the ball is:

a) 9.8N, 30◦ below horizontal
b) zero

c) 9.8N, up
d) 1.5N, down

124. A 5-kg block is suspended by a rope from the ceiling of an elevator as the elevator accelerates A 5-kg block is suspended by a rope from the ceiling of an elevator as the elevator accelerates downward at 3.0m/s2. The tension force of the rope on the block is:

a) 15 N, up
b) 34 N, up
c) 34 N, down
d) 64 N, up

125. You stand on a spring scale on the floor of an elevator. Of the following, the scale shows the highest reading when the elevator:

a) moves upward with increasing speed
b) moves upward with decreasing speed
c) remains stationary
d) moves downward with increasing speed

126. A block slides down a frictionless plane that makes an angle of 30◦ with the horizontal. The acceleration of the block is:

a) 980cm/s2
b) 566cm/s2
c) 849cm/s2
d) 490cm/s2

127. A 25-N crate is held at rest on a frictionless incline by a force that is parallel to the incline. If the incline is 25◦ above the horizontal the magnitude of the applied force is:

a) 4.1N
b) 4.6N
c) 8.9N
d) 11N

128. When a 40-N force, parallel to the incline and directed up the incline, is applied to a crate on a frictionless incline that is 30◦ above the horizontal, the acceleration of the crate is 2.0m/s2, up the incline. The mass of the crate is:

a) 3.8kg
b) 4.1kg
c) 5.8kg
d) 6.2kg

129. The "reaction" force does not cancel the "action" force because:

a) the action force is greater than the reaction force

b) they are on different bodies
c) they are in the same direction
d) the reaction force exists only after the action force is removed

130. A lead block is suspended from your hand by a string. The reaction to the force of gravity on the block is the force exerted by:

a) the string on the block
b) the block on the string
c) the string on the hand
d) the block on Earth

131. A brick slides on a horizontal surface. Which of the following will increase the magnitude of the frictional force on it?

a) Putting a second brick on top
b) Decreasing the surface area of contact
c) Increasing the surface area of contact
d) Decreasing the mass of the brick

132. A forward horizontal force of 12N is used to pull a 240-N crate at constant velocity across a horizontal floor. The coefficient of friction is:

a) 0.5
b) 0.05
c) 2
d) 0.2

133. A horizontal shove of at least 200N is required to start moving a 800-N crate initially at rest on a horizontal floor. The coefficient of static friction is:

a) 0.25
b) 0.125
c) 0.50
d) 4.00

134. A 12-kg crate rests on a horizontal surface and a boy pulls on it with a force that is 30◦ below the horizontal. If the coefficient of static friction is 0.40, the minimum magnitude force he needs to start the crate moving is:

a) 44N
b) 47N
c) 54N
d) 71N

135. A horizontal force of 5.0N pushes a 0.50-kg block against a vertical wall. The block is initially at rest. If μs =0.60 and μk =0.80, the acceleration of the block in m/s2 is:

a) 0
b) 1.8
c) 6.0
d) 8.0

136. A block is placed on a rough wooden plane. It is found that when the plane is tilted 30◦ to horizontal, the block will slide down at constant speed. The coefficient of kinetic friction of the block with the plane is:

a) 0.500
b) 0.577
c) 1.73
d) 0.866

137. A block is placed on a rough wooden plane. It is found that when the plane is tilted 30◦ to the horizontal, the block will slide down at constant speed. The coefficient of kinetic friction of the block with the plane is:

a) 0.500
b) 0.577
c) 1.73
d) 0.866

138. A 1000-kg airplane moves in straight flight at constant speed. The force of air friction is 1800 N. The net force on the plane is:

a) zero
b) 11800N
c) 1800N
d) 9800N

139. A ball is thrown upward into the air with a speed that is greater than terminal speed. On the way up it slows down and, after its speed equals the terminal speed but before it gets to the top of its trajectory:

a) its speed is constant
b) it continues to slow down
c) it speeds up
d) its motion becomes jerky

140. An object moving in a circle at constant speed:

a) must have only one force acting on it
b) is not accelerating
c) is held to its path by centrifugal force
d) has an acceleration of constant magnitude

141. An object moves around a circle. If the radius is doubled keeping the speed the same then the magnitude of the centripetal force must be:

a) twice as great
b) half as great
c) four times as great
d) one-fourth as great

142. A 800-N passenger in a car presses against the car door with a 200N force when the car makes a left turn at 13m/s. The (faulty) door will pop open under a force of 800 N. Of the following, the least speed for which the passenger is thrown out of the car is:

a) 14m/s
b) 19m/s
c) 20m/s
d) 26m/s

143. Which of the following is NOT a correct unit for work?

a) erg
b) ft·lb
c) watt
d) newton·meter

144. A boy holds a 40-N weight at arm's length for 10 s. His arm is 1.5m above the ground. The work done by the force of the boy on the weight while he is holding it is:

a) 0
b) 6.1J
c) 40 J
d) 60 J

145. An object moves in a circle at constant speed. The work done by the centripetal force is zero because:

a) the displacement for each revolution is zero
b) the average force for each revolution is zero
c) the magnitude of the acceleration is zero
d) the centripetal force is perpendicular to the velocity

146. An object of mass 1 g is whirled in a horizontal circle of radius 0.5m at a constant speed of 2m/s. The work done on the object during one revolution is:

a) 0
b) 1 J
c) 2 J
d) 4 J

147. A line drive to the shortstop is caught at the same height as it was originally hit. Over its entire flight the work done by gravity and the work done by air resistance, respectively, are:

a) zero; positive
b) zero; negative
c) positive; negative
d) negative; positive

148. A 2-kg object is moving at 3m/s. A 4-N force is applied in the direction of motion and then removed after the object has traveled an additional 5m. The work done by this force is:

a) 12 J
b) 15 J
c) 18 J
d) 20 J

149. A 0.50-kg object moves in a horizontal circular track with a radius of 2.5m. An external force of 3.0N, always tangent to the track, causes the object to speed up as it goes around. The work done by the external force as the mass makes one revolution is:

a) 24 J
b) 47 J
c) 59 J
d) 94 J

150. An 80-N crate slides with constant speed a distance of 5.0m downward along a rough slope that makes an angle of 30◦ with the horizontal. The work done by the force of gravity is:

a) −400 J
b) −200 J
c) −69 J
d) 200 J

151. n ideal spring is hung vertically from the ceiling. When a 2.0-kg block hangs at rest from it the spring is extended 6.0 cm from its relaxed length. A upward external force is then applied to the block to move it upward a distance of 16cm. While the block is moving upward the work done by the spring is:

a) −1.0J
b) −0.52 J
c) −0.26 J
d) 0.52 J

152. Which of the following bodies has the largest kinetic energy?

a) Mass 3M and speed V
b) Mass 3M and speed 2V
c) Mass 2M and speed 3V
d) Mass M and speed 4V

153. Which of the following is the correct combination of dimensions for energy?

a) MLT
b) LT2/m
c) ML2/T2
d) M2L3T

154. A 0.50-kg object moves on a horizontal frictionless circular track with a radius of 2.5m. An external force of 3.0N, always tangent to the track, causes the object to speed up as it goes around. If it starts from rest, then at the end of one revolution the radial component of the force of the track on it is:

a) 19N
b) 38N
c) 47N
d) 75N

155. In raising an object to a given height by means of an inclined plane, as compared with raising the object vertically, there is a reduction in:

a) work required
b) distance pushed
c) friction
d) force required

156. Only if a force on a particle is conservative:

a) is its work zero when the particle moves exactly once around any closed path
b) is its work always equal to the change in the kinetic energy of the particle
c) does it obey Newton's second law
d) does it obey Newton's third law

157. Two particles interact by conservative forces. In addition, an external force acts on each particle. They complete round trips, ending at the points where they started. Which of the following must have the same values at the beginning and end of this trip?

a) the total kinetic energy of the two-particle system
b) the potential energy of the two-particle system
c) the mechanical energy of the two-particle system
d) the total linear momentum of the two-particle system

158. A good example of kinetic energy is provided by:

a) a wound clock spring
b) the raised weights of a grandfather's clock
c) a tornado
d) a gallon of gasoline

159. Which one of the following five quantities CANNOT be used as a unit of potential energy?

a) watt·second
b) gram·cm/s2
c) gram·cm/s2 C. joule
d) kg·m2/s2

160. A 6.0-kg block is released from rest 80m above the ground. When it has fallen 60m its kinetic energy is approximately:

a) 4800 J
b) 3500 J
c) 1200 J
d) 120 J

161. A 2-kg block is thrown upward from a point 20m above Earth's surface. At what height above Earth's surface will the gravitational potential energy of the Earth-block system have increased by 500 J?

a) 5m
b) 25m
c) 46m
d) 70m

162. A projectile of mass 0.50 kg is fired with an initial speed of 10m/s at an angle of 60◦ above the horizontal. The potential energy of the projectile-Earth system (relative potential energy when the projectile is at ground level) is:

a) 25 J
b) 18.75 J
c) 12.5J
d) 6.25 J

163. A 2.2-kg block starts from rest on a rough inclined plane that makes an angle of 25◦ with the horizontal. The coefficient of kinetic friction is 0.25. As the block goes 2.0m down the plane, the mechanical energy of the Earth-block system changes by:

a) 0
b) −9.8J
c) 9.8J
d) −18 J

164. A force of 10N holds an ideal spring with a 20-N/m spring constant in compression. The potential energy stored in the spring is:

a) 0.5J
b) 2.5J
c) 5 J
d) 10 J

165. A 0.5-kg block slides along a horizontal frictionless surface at 2m/s. It is brought to rest by compressing a very long spring of spring constant 800N/m. The maximum spring compression is:

a) 0
b) 3 cm
c) 5 cm
d) 80cm

166. The center of mass of the system consisting of Earth, the Sun, and the planet Mars is:

a) closer to Earth than to either of the other bodies
b) closer to the Sun than to either of the other bodies
c) closer to Mars than to either of the other bodies
d) at the geometric center of the triangle formed by the three bodies

167. The center of mass of Earth's atmosphere is:

a) a little less than halfway between Earth's surface and the outer boundary of the atmosphere
b) near the surface of Earth
c) near the outer boundary of the atmosphere
d) near the center of Earth

168. Block A, with a mass of 4 kg, is moving with a speed of 2.0m/s while block B, with a mass of 8 kg, is moving in the opposite direction with a speed of 3m/s. The center of mass of the two block-system is moving with a velocity of:

a) 1.3m/s in the same direction as A
b) 1.3m/s in the same direction as B
c) 2.7m/s in the same direction as A
d) 1.0m/s in the same direction as B

169. Two 4.0-kg blocks are tied together with a compressed spring between them. They are thrown from the ground with an initial velocity of 35m/s, 45◦ above the horizontal. At the highest point of the trajectory they become untied and spring apart. About how far below the highest point is the center of mass of the two-block system 2.0 s later, before either fragment has hit the ground?

a) 12m
b) 20m
c) 31m
d) Can't tell because the velocities of the fragments are not given.

170. A 640-N hunter gets a rope around a 3200-N polar bear. They are stationary, 20m apart, on frictionless level ice. When the hunter pulls the polar bear to him, the polar bear will move:

a) 1.0m

b) 3.3m
c) 10m
d) 12m

171. A 2.0-kg mass is attached to one end of a spring with a spring constant of 100N/m and a 4.0-kg mass is attached to the other end. The masses are placed on a horizontal frictionless surface and the spring is compressed 10cm. The spring is then released with the masses at rest and the masses oscillate. When the spring has its equilibrium length for the first time the 2.0-kg mass has a speed of 0.36m/s. The mechanical energy that has been lost to the instant is:

a) zero
b) 0.31 J
c) 0.61 J
d) 0.81 J

172. Momentum may be expressed in:

a) kg/m
b) gram·s
c) N·s
d) kg/(m·s)

173. A rocket exhausts fuel with a velocity of 1500m/s, relative to the rocket. It starts from rest in outer space with fuel comprising 80 per cent of the total mass. When all the fuel has been exhausted its speed is:

a) 3600m/s
b) 2400m/s
c) 1200m/s
d) 880m/s

174. The physical quantity "impulse" has the same dimensions as that of:

a) force
b) power
c) energy
d) momentum

175. A 0.2-kg rubber ball is dropped from the window of a building. It strikes the sidewalk below at 30m/s and rebounds up at 20m/s. The impulse on the ball during the collision is:

a) 10N · s upward
b) 10N · s downward
c) 2.0N · s upward
d) 2.0N · s downward

176. A very massive object traveling at 10m/s strikes a very light object, initially at rest, and the light object moves off in the direction of travel of the heavy object. If the collision is elastic, the speed of the lighter object is:

a) 5.0m/s
b) 10m/s
c) 15m/s
d) 20m/s

177. Two objects, X and Y, are held at rest on a horizontal frictionless surface and a spring is compressed between them. The mass of X is 2/5 times the mass of Y. Immediately after the spring is released, X has a kinetic energy of 50 J and Y has a kinetic energy of:

a) 20 J
b) 8 J
c) 310 J
d) 125 J

178. A radian is about:

a) 25◦
b) 37◦
c) 45◦
d) 57◦

179. One revolution is the same as:

a) 1 rad
b) π/2 rad
c) π rad
d) 2π rad

180. The angular speed of the minute hand of a watch is:

a) (60/π)m/s
b) (1800/π)m/s
c) (π)m/s
d) (π/1800)m/s

181. A flywheel is initially rotating at 20 rad/s and has a constant angular acceleration. After 9.0s it has rotated through 450 rad. Its angular acceleration is:

a) 3.3 rad/s
b) 4.4 rad/s
c) 5.6 rad/s
d) 6.7 rad/s

182. A wheel initially has an angular velocity of 18 rad/s but it is slowing at a rate of 2.0 rad/s2. By the time it stops it will have turned through:

- **a) 81 rad**
- b) 160 rad
- c) 245 rad
- d) 330 rad

183. A wheel starts from rest and has an angular acceleration that is given by α(t)= (6.0 rad/s4)t2. The time it takes to make 10 rev is:

- a) 2.8s
- **b) 3.3s**
- c) 4.0s
- d) 4.7s

184. The angular velocity vector of a spinning body points out of the page. If the angular acceleration vector points into the page then:

- **a) the body is slowing down**
- b) the body is speeding up
- c) the body is starting to turn in the opposite direction
- d) the axis of rotation is changing orientation

185. The rotational inertia of a wheel about its axle does not depend upon its:

- a) diameter
- b) mass
- c) distribution of mass
- **d) speed of rotation**

186. A uniform solid cylinder made of lead has the same mass and the same length as a uniform solid cylinder made of wood. The rotational inertia of the lead cylinder compared to the wooden one is:

- a) greater
- **b) less**
- c) same
- d) unknown unless the radii are given

187. A disk is free to rotate on a fixed axis. A force of given magnitude F, in the plane of the disk, is to be applied. Of the following alternatives the greatest angular acceleration is obtained if the force is:

- a) applied tangentially halfway between the axis and the rim
- **b) applied tangentially at the rim**
- c) applied radially halfway between the axis and the rim
- d) applied radially at the rim

188. A thin circular hoop of mass 1.0 kg and radius 2.0m is rotating about an axis through its center and perpendicular to its plane. It is slowing down at the rate of 7.0 rad/s2. The net torque acting on it is:

a) 7.0N · m
b) 14.0N · m
c) 28.0N · m
d) 44.0N · m

189. A 0.70-kg disk with a rotational inertia given by MR2/2 is free to rotate on a fixed horizontal axis suspended from the ceiling. A string is wrapped around the disk and a 2.0-kg mass hangs from the free end. If the string does not slip, then as the mass falls and the cylinder rotates, the suspension holding the cylinder pulls up on the cylinder with a force of:

a) 6.9N
b) 9.8N
c) 16N
d) 26N

190. A disk starts from rest and rotates around a fixed axis, subject to a constant net torque. The as the work done during the first 5 s. work done by the torque during the second 5 s is

a) the same
b) twice as much
c) half as much
d) four times as much

191. A forward force on the axle accelerates a rolling wheel on a horizontal surface. If the wheel does not slide the frictional force of the surface on the wheel is:

a) zero
b) in the forward direction
c) in the backward direction
d) in the upward direction

192. The coefficient of static friction between a certain cylinder and a horizontal floor is 0.40. If the rotational inertia of the cylinder about its symmetry axis is given by I =(1/2)MR2, then the magnitude of the maximum acceleration the cylinder can have without sliding is:

a) 0.1g
b) 0.2g
c) 0.4g
d) 0.8g

193. A hoop, a uniform disk, and a uniform sphere, all with the same mass and outer radius, start with the same speed and roll without sliding up identical inclines. Rank the objects according to how high they go, least to greatest.

a) hoop, disk, sphere

b) disk, hoop, sphere
c) sphere, hoop, disk
d) sphere, disk, hoop

194. When we apply the energy conservation principle to a cylinder rolling down an incline without sliding, we exclude the work done by friction because:

a) there is no friction present
b) the angular velocity of the center of mass about the point of contact is zero
c) the coefficient of kinetic friction is zero
d) the linear velocity of the point of contact (relative to the inclined surface) is zero

195. A 5.0-kg ball rolls without sliding from rest down an inclined plane. A 4.0-kg block, mounted on roller bearings totaling 100 g, rolls from rest down the same plane. At the bottom, the block has:

a) greater speed than the ball
b) less speed than the ball
c) the same speed as the ball
d) greater or less speed than the ball, depending on the angle of inclination

196. Possible units of angular momentum are:

a) kg·m/s
b) kg·m2/s2
c) kg·m/s2
d) kg·m2/s

197. The newton·second is a unit of:

a) work
b) angular momentum
c) power
d) linear momentum

198. A 2.0-kg stone is tied to a 0.50-m long string and swung around a circle at a constant angular velocity of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, 0.75m from the origin. The magnitude of the torque about the origin is:

a) 0
b) 6.0N · m
c) 14N · m
d) 108N · m

199. A man, holding a weight in each hand, stands at the center of a horizontal frictionless rotating turntable. The effect of the weights is to double the rotational inertia of the system. As he is rotating, the man opens his hands and drops the two weights. They fall outside the turntable. Then:

a) his angular velocity doubles
b) his angular velocity remains about the same
c) his angular velocity is halved
d) the direction of his angular momentum vector changes

200. A net torque applied to a rigid object always tends to produce:

a) linear acceleration
b) rotational equilibrium
c) angular acceleration
d) rotational inertia

201. For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

a) the center of mass
b) the center of gravity
c) the geometrical center
d) the same point

202. To determine if a rigid body is in equilibrium the vector sum of the gravitational forces acting on the particles of the body can be replaced by a single force acting at:

a) the center of mass
b) the geometrical center
c) the center of gravity
d) a point on the boundary

203. The center of gravity coincides with the center of mass:

a) always
b) never
c) if the center of mass is at the geometrical center of the body
d) if the acceleration due to gravity is uniform over the body

204. The location of which of the following points within an object might depend on the orientation of the object?

a) Its center of mass
b) Its center of gravity
c) Its geometrical center
d) Its center of momentum

205. A 160-N child sits on a light swing and is pulled back and held with a horizontal force of 100 N. The magnitude of the tension force of each of the two supporting ropes is:

a) 60N
b) 94N

c) 120N
d) 190N

206. A uniform plank is 6.0m long and weighs 80 N. It is balanced on a sawhorse at its center. An additional 160N weight is now placed on the left end of the plank. To keep the plank balanced, it must be moved what distance to the left?

a) 6.0m
b) 2.0m
c) 1.5m
d) 1.0m

207. A uniform 240-g meter stick can be balanced by a 240-g weight placed at the 100-cm mark if the fulcrum is placed at the point marked:

a) 75cm
b) 60cm
c) 50cm
d) 40cm

208. A window washer attempts to lean a ladder against a frictionless wall. He finds that the ladder slips on the ground when it is placed at an angle of less than 75◦ to the ground but remains in place when the angle is greater than 75◦. The coefficient of static friction between the ladder and the ground:

a) is about 0.13
b) is about 0.27
c) is about 1.0
d) depends on the mass of the ladder

209. Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:

a) the shear
b) the fractional change in volume
c) the fractional change in length
d) the pressure

210. The ultimate strength of a sample is the stress at which the sample:

a) returns to its original shape when the stress is removed
b) remains underwater
c) breaks
d) bends 180◦

211. The bulk modulus is a proportionality constant that relates the pressure acting on an object to:

a) the shear

b) the fractional change in volume
c) the fractional change in length
d) Young's modulus

212. A cube with 2.0-cm sides is made of material with a bulk modulus of 4.7 × 105 N/m2. When it is subjected to a pressure of 2.0 × 105 Pa the length of its any of its sides is:

a) 0.85cm
b) 1.15cm
c) 1.66cm
d) 2.0cm

214. Let F1 be the magnitude of the gravitational force exerted on the Sun by Earth and F2 be the magnitude of the force exerted on Earth by the Sun. Then:

a) F1 is much greater than F2
b) F1 is slightly greater than F2
c) F1 is equal to F2
d) F1 is slightly less than F2

215. Venus has a mass of about 0.0558 times the mass of Earth and a diameter of about 0.381 times the diameter of Earth. The acceleration of a body falling near the surface of Venus is about:

a) 0.21m/s2
b) 1.4m/s2
c) 2.8m/s2
d) 3.8m/s2

216. The mass of an object:

a) is slightly different at different locations on Earth
b) is a vector
c) is independent of the acceleration due to gravity
d) is the same for all objects of the same size and shape

217. The mass of a hypothetical planet is 1/100 that of Earth and its radius is 1/4 that of Earth. If a person weighs 600N on Earth, what would he weigh on this planet?

a) 24N
b) 48N
c) 96N
d) 192N

218. A spring scale, calibrated in newtons, is used to weigh sugar. If it were possible to weigh sugar at the following locations, where will the buyer get the most sugar to a newton?

a) At the north pole

b) At the equator
c) At the center of Earth
d) On the Moon

219. An astronaut in an orbiting spacecraft feels "weightless" because she:

a) is beyond the range of gravity
b) is pulled outward by centrifugal force
c) has no acceleration
d) has the same acceleration as the spacecraft

220. In planetary motion the line from the star to the planet sweeps out equal areas in equal times. This is a direct consequence of:

a) the conservation of energy
b) the conservation of momentum
c) the conservation of angular momentum
d) the conservation of mass

221. A planet is in circular orbit around the Sun. Its distance from the Sun is four times the average distance of Earth from the Sun. The period of this planet, in Earth years, is:

a) 4
b) 8
c) 16
d) 64

222. An artificial satellite of Earth nears the end of its life due to air resistance. While still in orbit:

a) it moves faster as the orbit lowers
b) it moves slower as the orbit lowers
c) it slowly spirals away from Earth
d) it moves slower in the same orbit but with a decreasing period

223. All fluids are:

a) gases
b) liquids
c) gases or liquids
d) non-metallic

224. Gases may be distinguished from other forms of matter by their:

a) lack of color
b) small atomic weights
c) inability to form free surfaces
d) ability to flow

225. Barometers and open-tube manometers are two instruments that are used to measure pressure.

a) Both measure gauge pressure
b) Both measure absolute pressure
c) Barometers measure gauge pressure and manometers measure absolute pressure
d) Barometers measure absolute pressure and manometers measure gauge pressure

226. The pressure exerted on the ground by a man is greatest when:

a) he stands with both feet flat on the ground
b) he stands flat on one foot
c) he stands on the toes of one foot
d) he lies down on the ground

227. In a stationary homogeneous liquid:

a) pressure is the same at all points
b) pressure depends on the direction
c) pressure is independent of any atmospheric pressure on the upper surface of the liquid
d) pressure is the same at all points at the same level

228. A uniform U-tube is partially filled with water. Oil, of density 0.75 g/cm3, is poured into the right arm until the water level in the left arm rises 3 cm. The length of the oil column is then:

a) 2.25cm
b) 8 cm
c) 6 cm
d) 4 cm

229. A student standardizes the concentration of a saltwater solution by slowly adding salt until an egg will just float. The procedure is based on the assumption that:

a) all eggs have the same volume
b) all eggs have the same weight
c) all eggs have the same density
d) all eggs have the same shape

230. A steel ax and an aluminum piston have the same apparent weight in water. When they are weighed in air:

a) they weigh the same
b) the ax is heavier
c) the piston is heavier
d) both weigh less than they did in water

231. The apparent weight of a steel sphere immersed in various liquids is measured using a spring scale. The greatest reading is obtained for that liquid:

a) having the smallest density
b) having the largest density
c) subject to the greatest atmospheric pressure
d) having the greatest volume

232. An object floats on the surface of a fluid. For purposes of calculating the torque on it, the buoyant force is taken to act at:

a) the center of the bottom surface of the object
b) the center of gravity of the object
c) the center of gravity of the fluid that the object replaced
d) the geometric center of the object

233. A blast of wind tips a sailboat in the clockwise direction when viewed from the stern. When the wind ceases the boat rotates back toward the upright position if, when it is tilted, the center of buoyancy:

a) is above the center of gravity
b) is below the center of gravity
c) is to the right of the center of gravity
d) is to the left of the center of gravity

234. Two balls have the same shape and size but one is denser than the other. If frictional forces are negligible when they are dropped in air, which has the greater acceleration?

a) The heavier ball
b) The lighter ball
c) They have the same acceleration
d) The heavier ball if atmospheric pressure is high, they lighter ball if it is low

235. The equation of continuity for fluid flow can be derived from the conservation of:

a) energy
b) mass
c) angular momentum
d) volume

236. Which of the following assumptions is NOT made in the derivation of Bernoulli's equation?

a) Assume streamline flow
b) Neglect viscosity
c) Neglect friction
d) Neglect gravity

237. A non-viscous incompressible liquid is flowing through a horizontal pipe of constant cross section. Bernoulli's equation and the equation of continuity predict that the drop in pressure along the pipe:

a) is zero
b) depends on the length of the pipe
c) depends on the fluid velocity
d) depends on the cross-sectional area of the pipe

238. A large tank filled with water has two holes in the bottom, one with twice the radius of the other. In steady flow the speed of water leaving the larger hole is________ the speed of the water leaving the smaller.

a) twice
b) four times
c) half
d) the same as

239. A person blows across the top of one arm of a U-tube partially filled with water. The water in that arm:

a) rises slightly
b) drops slightly
c) remains at the same height
d) rises if the blowing is soft but drops if it is hard

240. In simple harmonic motion, the restoring force must be proportional to the:

a) amplitude
b) frequency
c) velocity
d) displacement

241. A particle is in simple harmonic motion with period T. At time t = 0 it is at the equilibrium point. Of the following times, at which time is it furthest from the equilibrium point?

a) 0.5T
b) 0.7T
c) T
d) 1.4T

242. An object is undergoing simple harmonic motion. Throughout a complete cycle it:

a) has constant speed
b) has varying amplitude
c) has varying period
d) has varying acceleration

243. An object attached to one end of a spring makes 20 complete oscillations in 10 s. Its period is:

a) 2Hz
b) 10 s
c) 0.5Hz
d) 0.50 s

244. A block attached to a spring oscillates in simple harmonic motion along the x axis. The limits of its motion are x = 10 cm and x = 50 cm and it goes from one of these extremes to the other in 0.25 s. Its amplitude and frequency are:

a) 40cm, 2 Hz
b) 20cm, 4 Hz
c) 40cm, 2 Hz
d) 25cm, 4 Hz

245. It is impossible for two particles, each executing simple harmonic motion, to remain in phase with each other if they have different:

a) masses
b) periods
c) amplitudes
d) spring constants

246. A 0.20-kg object attached to a spring whose spring constant is 500N/m executes simple har-monic motion. If its maximum speed is 5.0m/s, the amplitude of its oscillation is:

a) 0.0020m
b) 0.10m
c) 0.20m
d) 25m

247. A particle moves in simple harmonic motion according to x = 2 cos(50t), where x is in meters and t is in seconds. Its maximum velocity in m/s is:

a) 100 sin(50t)
b) 100 cos(50t)
c) 100
d) 200

248. A block attached to a spring undergoes simple harmonic motion on a horizontal frictionless surface. Its total energy is 50 J. When the displacement is half the amplitude, the kinetic energy is:

a) zero
b) 12.5J
c) 25 J
d) 37.5J

249. A simple pendulum consists of a small ball tied to a string and set in oscillation. As the pendulum swings the tension force of the string is:

a) constant
b) a sinusoidal function of time
c) the square of a sinusoidal function of time
d) None of the above

250. The rotational inertia of a uniform thin rod about its end is ML2/3, where M is the mass and L is the length. Such a rod is hung vertically from one end and set into small amplitude oscillation. If L =1.0m this rod will have the same period as a simple pendulum of length:

a) 33cm
b) 50cm
c) 67cm
d) 100cm

251. A sinusoidal force with a given amplitude is applied to an oscillator. To maintain the largest amplitude oscillation the frequency of the applied force should be:

a) half the natural frequency of the oscillator
b) the same as the natural frequency of the oscillator
c) twice the natural frequency of the oscillator
d) unrelated to the natural frequency of the oscillator

252. Water waves in the sea are observed to have a wavelength of 300m and a frequency of 0.07 Hz. The speed of these waves is:

a) 0.00021m/s
b) 2.1m/s
c) 21m/s
d) 210m/s

253. A transverse traveling sinusoidal wave on a string has a frequency of 100 Hz, a wavelength of 0.040 m, and an amplitude of 2.0mm. The maximum velocity in m/s of any point on the string is:

a) 0.2
b) 1.3
c) 4
d) 15

254. The tension in a string with a linear mass density of 0.0010 kg/mis 0.40 N. A sinusoidal wave with a wavelength of 20cm on this string has a frequency of:

a) 0.0125 Hz
b) 0.25 Hz
c) 100 Hz

d) 630 Hz

255. A wave on a string is reflected from a fixed end. The reflected wave:

a) is in phase with the original wave at the end
b) is 180◦ out of phase with the original wave at the end
c) has a larger amplitude than the original wave
d) has a larger speed than the original wave

256. A string of length L is clamped at each end and vibrates in a standing wave pattern. The wavelengths of the constituent traveling waves CANNOT be:

a) L
b) 2L
c) L/2
d) 4L

257. Two sinusoidal waves, each of wavelength 5m and amplitude 10cm, travel in opposite directions on a 20-m long stretched string that is clamped at each end. Excluding the nodes at the ends of the string, how many nodes appear in the resulting standing wave?

a) 3
b) 4
c) 5
d) 7

258. When a string is vibrating in a standing wave pattern the power transmitted across an antinode, compared to the power transmitted across a node, is:

a) more
b) less
c) the same (zero)
d) the same (non-zero)

259. The speed of a sound wave is determined by:

a) its amplitude
b) its intensity
c) its pitch
d) the transmitting medium

260. A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:

a) 0.75m
b) 1.5m
c) 3.0m
d) need to know wave speed

261. At points in a sound wave where the gas is maximally compressed, the pressure

a) is a maximum
b) is a minimum
c) is equal to the ambient value
d) is greater than the ambient value but less than the maximum

262. "Beats" in sound refer to:

a) interference of two waves of the same frequency
b) combination of two waves of slightly different frequency
c) reversal of phase of reflected wave relative to incident wave
d) two media having slightly different sound velocities

263. The sound intensity 5.0m from a point source is 0.50W/m2. The power output of the source is:

a) 39W
b) 160W
c) 266W
d) 320W

264. The intensity of a certain sound wave is 6μW/cm2. If its intensity is raised by 10 db, the new intensity (in μW/cm2) is:

a) 60
b) 6.6
c) 6.06
d) 600

265. The sound level at a point P is 14 db below the sound level at a point 1.0m from a point source. The distance from the source to point P is:

a) 4.0cm
b) 202m
c) 2.0m
d) 5.0m

266. If the length of a piano wire (of given density) is increased by 5%, what approximate change in tension is necessary to keep its fundamental frequency unchanged?

a) Decrease of 10%
b) Decrease of 5%
c) Increase of 5%
d) Increase of 10%

267. A piano wire has a length of 81cm and a mass of 2.0 g. If its fundamental frequency is to be 394 Hz, its tension must be:

a) 0.32N
b) 63N
c) 130N
d) 250N

268. A column of argon is open at one end and closed at the other. The shortest length of such a column that will resonate with a 200 Hz tuning fork is 42.5 cm. The speed of sound in argon must be:

a) 85.0m/s
b) 170m/s
c) 340m/s
d) 470m/s

269. An organ pipe with one end open and the other closed is operating at one of its resonant frequencies. The open and closed ends are respectively:

a) pressure node, pressure node
b) pressure node, displacement node
c) displacement antinode, pressure node
d) displacement node, displacement node

270. If the speed of sound is 340m/s, the length of the shortest closed pipe that resonates at 218 Hz is:

a) 23cm
b) 17cm
c) 39cm
d) 78cm

271. If the speed of sound is 340m/s, the two lowest frequencies of an 0.5-m organ pipe, closed at one end, are approximately:

a) 170 and 340 Hz
b) 170 and 510 Hz
c) 340 and 680 Hz
d) 340 and 1020 Hz

272. The rise in pitch of an approaching siren is an apparent increase in its:

a) speed
b) amplitude
c) frequency
d) wavelength

273. A source emits sound with a frequency of 1000 Hz. It and an observer are moving in the same direction with the same speed, 100m/s. If the speed of sound is 340m/s, the observer hears sound with a frequency of:

a) 294 Hz
b) 545 Hz
c) 1000 Hz
d) 1830 Hz

274. If two objects are in thermal equilibrium with each other:

a) they cannot be moving
b) they cannot be undergoing an elastic collision
c) they cannot have different pressures
d) they cannot be at different temperatures

275. A balloon is filled with cold air and placed in a warm room. It is NOT in thermal equilibrium with the air of the room until:

a) it rises to the ceiling
b) it sinks to the floor
c) it stops expanding
d) it starts to contract

276. A constant-volume gas thermometer is used to measure the temperature of an object. When the thermometer is in contact with water at its triple point (273.16 K) the pressure in the thermometer is 8.500 × 104 Pa. When it is in contact with the object the pressure is 9.650 × 104 Pa. The temperature of the object is:

a) 37.0K
b) 241K
c) 310K
d) 314K

277. Fahrenheit and Kelvin scales agree numerically at a reading of:

a) -40
b) 0
c) 273
d) 574

278. Room temperature is about 20 degrees on the:

a) Kelvin scale
b) Celsius scale
c) Fahrenheit scale
d) absolute scale

279. The air temperature on a summer day might be about:

a) 0◦ C
b) 10◦ C
c) 25◦ C
d) 80◦ C

280. It is more difficult to measure the coefficient of volume expansion of a liquid than that of a solid because:

a) no relation exists between linear and volume expansion coefficients
b) a liquid tends to evaporate
c) a liquid expands too much when heated
d) the containing vessel also expands

281. The coefficient of linear expansion of steel is 11 × 10−6 per C◦ . A steel ball has a volume of exactly 100cm3 at 0◦ C. When heated to 100◦ C, its volume becomes:

a) 100.33cm3
b) 100.0011cm3
c) 100.0033cm3
d) 100.000011cm3

282. Metal pipes, used to carry water, sometimes burst in the winter because:

a) metal contracts more than water
b) outside of the pipe contracts more than the inside
c) metal becomes brittle when cold
d) water expands when it freezes

283. Heat is:

a) energy transferred by virtue of a temperature difference
b) energy transferred by macroscopic work
c) energy content of an object
d) a temperature difference

284. Heat has the same units as:

a) temperature
b) work
c) energy/time
d) heat capacity

285. A cube of aluminum has an edge length of 20cm. Aluminum has a density 2.7 times that of water (1 g/cm3) and a specific heat 0.217 times that of water (1 cal/g · C◦). When the internal energy of the cube increases by 47000 cal its temperature increases by:

a) 5C◦

b) 10C◦
c) 20C◦
d) 100C◦

286. A heat of transformation of a substance is:

a) the energy absorbed as heat during a phase transformation
b) the energy per unit mass absorbed as heat during a phase transformation
c) the same as the heat capacity
d) the same as the specific heat

287. Evidence that a gas consists mostly of empty space is the fact that:

a) the density of a gas becomes much greater when it is liquefied
b) gases exert pressure on the walls of their containers
c) gases are transparent
d) heating a gas increases the molecular motion

288. It is known that 28 g of a certain ideal gas occupy 22.4 liters at standard conditions (0◦ C, 1 atm). The volume occupied by 42 g of this gas at standard conditions is:

a) 14.9 liters
b) 22.4 liters
c) 33.6 liters
d) 42 liters

289. A 2-m3 weather balloon is loosely filled with helium at 1 atm (76cm Hg) and at 27◦ C. At an elevation of 20, 000 ft, the atmospheric pressure is down to 38cm Hg and the helium has expanded, being under no constraint from the confining bag. If the temperature at this elevation is -48◦ C, the gas volume (in m3) is:

a) 3
b) 4
c) 2
d) 2.5

290. The temperature of low pressure hydrogen is reduced from 100◦ Cto 20◦ C. The rms speed of its molecules decreases by approximately:

a) 80%
b) 89%
c) 46%
d) 11%

291. The principle of equipartition of energy states that the internal energy of a gas is shared equally:

a) among the molecules

b) between kinetic and potential energy
c) among the relevant degrees of freedom
d) between translational and vibrational kinetic energy

292. The number of degrees of freedom of a triatomic molecule is:

a) 1
b) 3
c) 6
d) 9

293. The internal energy of an ideal gas depends on:

a) the temperature only
b) the pressure only
c) the volume only
d) the temperature and pressure only

294. The pressure of an ideal gas of diatomic molecules is doubled by halving the volume. The ratio of the new internal energy to the old, both measured relative to the internal energy at 0K, is:

a) 1/4
b) 1/2
c) 1
d) 2

295. When work W is done on an ideal gas of diatomic molecules in thermal isolation the increase in the total rotational energy of the molecules is:

a) 0
b) W/3
c) 2W/3
d) 2W/5

296. The Maxwellian speed distribution provides a direct explanation of:

a) thermal expansion
b) the ideal gas law
c) heat
d) evaporation

297. In a reversible process the system:

a) is always close to equilibrium states
b) is close to equilibrium states only at the beginning and end
c) might never be close to any equilibrium state
d) is close to equilibrium states throughout, except at the beginning and end

298. A heat engine that in each cycle does positive work and loses energy as heat, with no heat energy input, would violate:

a) the zeroth law of thermodynamics
b) the first law of thermodynamics
c) the second law of thermodynamics
d) the third law of thermodynamics

299. A coulomb is the same as:

a) an ampere/second
b) half an ampere·second2
c) an ampere/meter2
d) an ampere·second

300. A conductor is distinguished from an insulator with the same number of atoms by the number of:

a) nearly free atoms
b) electrons
c) nearly free electrons
d) protons

301. An electric field is most directly related to:

a) the momentum of a test charge
b) the kinetic energy of a test charge
c) the potential energy of a test charge
d) the force acting on a test charge

302. The torque exerted by an electric field on a dipole is:

a) parallel to the field and perpendicular to the dipole moment
b) parallel to both the field and dipole moment
c) perpendicular to both the field and dipole moment
d) parallel to the dipole moment and perpendicular to the field